Effect of Remittances on Farm Productivity after a Large Natural Disaster [*]

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We study how remittance responses to a large natural disaster affect household production. We combine pre- and post-earthquake household survey data from Nepal with spatial variation in earthquake intensity and exchange-rate-induced remittance variation in a triple-difference instrumental-variables design. We find that higher remittance inflows in high-intensity earthquake areas reduced household production in the short run, while total consumption fell by an order of magnitude less. In this sense, remittances appear to crowd out household production. Households adjust primarily through lower capital inputs, lower agricultural labor, and lower production in relatively capital-intensive activities. We find no evidence that public assistance moves with remittance inflows, so the adjustment does not appear to run through substitution between private and public transfers. From a policy perspective, our results suggest that remittances may have helped households cope with large shocks, but they may also reduce the need to maintain production when other post-disaster needs become more pressing.

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1 Introduction

For many low- and middle-income countries, migration income is a core part of how households finance consumption, cope with risk, and make production decisions. Remittance inflows to these economies reached $651 billion in 2022 alone (Ratha et al. 2024). Nepal is a particularly useful setting in which to study these effects because it is highly dependent on both remittances and agriculture: 22% of households have at least one member living abroad, while agriculture, forestry, and fishing remain the main production sector for 64% of the working-age population (NSO 2023).[1] Remittances are also among Nepal's largest external resource flows, accounting for more than 20% of GDP in recent years (Figure 1). Remittances and farm production are therefore jointly central to household welfare in Nepal.

Much of the existing literature studies how migration and remittances shape origin-household welfare, investment, and labor allocation in normal times (Mobarak, Sharif, and Shrestha 2023; Yang 2008b; Mishra, Kondratjeva, and Shively 2022). More recent work also shows that exchange-rate-induced changes in migrant income opportunities can have persistent effects on origin-area development (Khanna et al. 2026). Much less is known, however, about how remittance inflows affect production choices when households are suddenly hit by a major shock. That question matters more as extreme events become more frequent and as climate change, demographic pressure, and migration reshape local economies (McAuliffe and Triandafyllidou 2021; CRED 2015; IPCC 2014). When disaster strikes, remittances may become more important precisely because households face urgent and competing demands on cash, labor, and assets.

This paper studies how the remittance response to a large disaster changes household production choices. We focus on the 2015 Nepal earthquake, which caused more than 9,000 deaths and displaced more than 500,000 households (Government of Nepal, National Planning Commission 2015). The earthquake created an immediate need for liquidity in affected areas. Extra remittances could, in principle, help households maintain production by financ-

1. These figures are from the 2021 National Population and Housing Census, the most recent comprehensive source. Patterns were similar around our estimation window: at the 2011 census, 25.4% of households reported at least one member absent abroad, and 55.8% of households received remittances in 2010/11 (CBS 2011).

ing agricultural inputs and labor. But they could also ease broader short-run pressure on the household budget, making it feasible for recipient households to cut back on farming when that option is less available to households without remittance support. In this sense, remittances may finance agricultural investment, but they may also reduce the need to devote time, labor, and working capital to production. Our question is therefore whether the remittance response to a large shock preserves production or instead allows households to reallocate away from it.

Identifying that effect is difficult because remittances are not randomly assigned. Migration opportunities, migration risks, and the returns to migration differ across households, so remittance receipt is closely linked to underlying household choices and constraints (Mobarak, Sharif, and Shrestha 2023; Theoharides 2017; Shrestha 2019). Remittances can also respond endogenously to adverse household shocks and distress, precisely because they often serve an insurance role (Yang and Choi 2007; Yang 2008a).

We find that higher remittance inflows in more severely affected areas reduced household production in the short run. The pattern is not easily explained by a simple adverse-selection story among remittance households. Instead, the estimates indicate much smaller adjustments in consumption than in production, alongside lower agricultural capital inputs and labor use, and lower production in categories with larger ongoing cash commitments. The mediation exercise points in the same direction: the production effect is linked more closely to changes in capital and labor than to changes in consumption or other financing margins.

These results are consistent with remittances easing short-run budget pressure after the shock and reducing the need to sustain farm activity at pre-earthquake levels. Remittance income may also affect labor-supply decisions within the household when there is an outside source of cash, a channel that is consistent with evidence from Nepal on left-behind household members (Lokshin and Glinskaya 2009; Phadera 2016). Our evidence does not isolate a single mechanism, but it points to post-disaster adjustment operating through reduced pressure to maintain agricultural production.

We contribute foremost to the literature on remittances and shocks. Existing work shows that remittances often rise when origin households are hit by adverse events and can stabilize external financial inflows after disasters (Yang and Choi 2007; Yang 2008a; Eldemerdash and

Landis 2023). Much less is known about what those additional remittances do to household production after a major shock. We provide causal evidence on that margin in the context of a large earthquake and show that larger remittance responses are associated with lower farm production, not stronger production recovery. In our setting, remittances relax the need to maintain agricultural activity rather than finance a return to pre-disaster production.

We also contribute to the literature on agricultural production and rural adjustment in developing countries. A large literature studies how relaxing credit and risk constraints can increase input use, technology adoption, and output (Karlan et al. 2014; Duflo, Kremer, and Robinson 2008). Our results highlight a different margin. After a severe disaster, additional liquidity from abroad need not translate into higher production. In our setting, the evidence instead points to lower input use and lower production, suggesting that remittances operate differently from interventions aimed directly at raising farm investment.

More specifically, we show that smoothing consumption does not imply preserving production. Prior work shows that remittances can insure recipient households against adverse income shocks and reduce consumption instability (Gubert 2002; Yang and Choi 2007; Combes and Ebeke 2011; Ebeke and Combes 2013). In our estimates, total consumption falls by far less than production. That combination indicates that post-disaster remittance responses can be consistent with consumption smoothing even when agricultural production contracts. Put differently, remittances appear to crowd out farm production in order to keep consumption smooth.

Finally, we provide evidence on mechanisms using a structured mediation exercise, following the logic of Imai, Keele, and Tingley (2010), in addition to the reduced-form decomposition. We examine whether the fall in production is more closely associated with changes in total consumption, capital inputs, labor inputs, or public assistance, one mediator at a time. The mediation results reinforce the broader pattern in the reduced-form tables: the decline in production is linked much more closely to lower capital and labor use than to lower consumption.

The remainder of the paper proceeds as follows. Section 2 describes the migration, remittance, and earthquake context in Nepal. Section 3 describes the household survey data and the construction of our main outcome and treatment variables. Section 4 presents the empirical strategy and results, including the motivating land-cover evidence, the triple-difference IV design and identification argument, the main production results, and the mechanism and mediation analysis. Section 5 concludes.

2 Background

2.1 Migration, Remittances, and Agriculture

Nepal is both an agrarian economy and a remittance-dependent economy. A large share of households depend directly or indirectly on agriculture, while labor migration has become a central part of household risk management and income diversification. Migration to India has long been important, but in recent decades migration to Gulf countries, Malaysia, Japan, and South Korea has expanded substantially (Seddon, Adhikari, and Gurung 2002; World Bank 2011; Shrestha 2019). As a result, remittances have become a major source of liquidity for households and an important macro-level source of foreign exchange earnings (Sapkota 2013). This combination makes Nepal a useful setting for studying how private transfers shape household adjustment after a large shock, because migration, remittances, and agriculture are all quantitatively central to rural livelihoods. Figure 1 illustrates this macro dependence by plotting remittances relative to GDP over time using our own calculations.

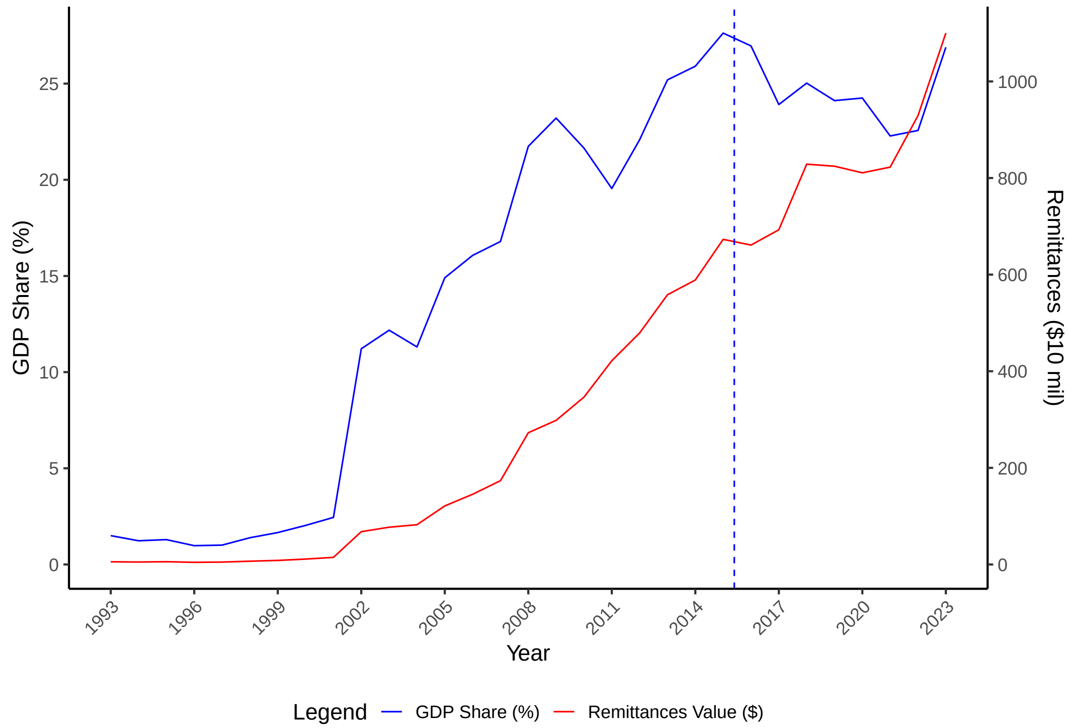

Figure 1 — Remittances Relative to GDP Over Time

Note: This figure plots remittances relative to GDP in Nepal over time. Source: Authors' calculations.

2.2 The 2015 Nepal Earthquake

The April 2015 earthquake was one of the deadliest natural disasters in Nepal's recent history. It caused extensive human and economic losses and left many households facing urgent spending needs and severe disruption (USGS 2015; Goda et al. 2015). In such a setting, remittances can act as private insurance. But private insurance does not necessarily preserve productive investment. If households face urgent post-disaster spending needs, remittances may stabilize welfare while expenditure on production inputs falls. Recent work on Nepal's earthquake shows that the shock generated large and persistent pressure on local economic and institutional outcomes (Pathak and Schündeln 2022, 2026). Our contribution is to bring that same post-disaster perspective to household remittance responses and agricultural production.

3 Data and Variables

3.1 Household Surveys

We combine the third round of the Nepal Living Standards Survey (NLSS, 2010/11) with the Household Risk and Vulnerability Survey (HRVS, 2016–2018) (CBS 2011; World Bank 2020). The NLSS is an independent sample and provides our pre-earthquake baseline, while the HRVS is a three-year post-earthquake panel that resurveys the same households in 2016, 2017 and 2018. Both are multi-topic household surveys covering the household roster, migration and remittances, agricultural production, consumption expenditure and household finance. Both also identify the ward where the household lives, so we can attach earthquake intensity to each household. Because the NLSS sample is independent, no household appears both before and after the earthquake, and we therefore compare households within a location rather than within a household.

The unit of observation is the household-year. We build the panel from the household rosters and the migration modules of both surveys, and we link each migrant back to their roster record wherever the survey reports a member identifier. Production, consumption and the input blocks are all household quantities, so the household-year is the level at which the outcome actually varies. Remittances belong to the individual migrant, and we aggregate them to the household by summing over its migrants. The exchange rate is aggregated the same way, as the simple average of the rates faced by the household's migrants, which is the migrant-count weighting used by Yang (2008b). A household with migrants in two countries therefore enters once, carrying the average of the two rates.

Our estimation sample contains 22,354 household-years, drawn from 11,460 distinct households in 875 wards, 712 VDCs and 71 districts. The 2011 baseline contributes 5,899 household-years, and the post-earthquake rounds contribute 5,515, 5,425 and 5,515, so the panel is close to balanced across the four rounds. Of the 22,354 household-years, 11,280 have at least one migrant and 7,271 have a migrant abroad; the latter are the household-years that carry exchange-rate variation.

3.2 Variables

Production. Our main outcome is household farm production, the value of crop and livestock output over the survey year. Crop production is built from the quantities the household reports harvesting of each crop, valued at the prices prevailing in its own local market as recorded in the survey. Livestock production is the value of livestock output rather than of livestock sales, so produce consumed inside the household still counts; it covers meat, dairy, eggs and other livestock produce. Total production sums ten crop categories (cereals, pulses, tubers, oilseed, cash crops, spices, vegetables, citrus and non-citrus fruit, and other crops) and four livestock categories. We convert all monetary variables in the paper to 2018 Nepalese rupees using the national consumer price index, so the baseline round and the three post-earthquake rounds are comparable.

Consumption, farm inputs and public assistance. We build three further outcome blocks the same way, as sums of their survey components. Consumption covers the food categories together with fuel, apparel, utilities, durables, housing improvement and ceremonial spending. Capital inputs cover the cash-intensive agricultural inputs: seeds, fertilizer, irrigation and rented equipment. Labor inputs cover spending on hired agricultural labor and the imputed value of the household's own agricultural labor. We also report total labor hours, which sum wage and self-employed hours inside and outside agriculture, in Table 1 and Appendix Table A5. Public assistance covers government and program transfers, including earthquake relief, the old-age and widow allowance, disability and ethnic-minority allowances, maternal and child grants and conflict-related payments. Appendix A.1 lists the components of each block.

Remittances. Remittances are the amount a migrant sent to the household over the survey year, which the household reports separately for each migrant. We sum over the household's migrants, so the measure is the household's total receipts and is zero for households with no migrant. A substantial share of households receive nothing, so the benchmark specification enters remittances as $\log(1+R)$, which keeps those households in the estimating sample. Appendix A.1 reports the inverse hyperbolic sine and levels alternatives.

Earthquake intensity. Earthquake intensity is the Modified Mercalli Intensity (MMI) at the household's ward, taken from the USGS ShakeMap for the 25 April 2015 event. MMI measures the ground shaking actually experienced at a location rather than the magnitude of the earthquake at its source, so it captures exposure rather than distance to the epicenter. We define HI_v as an indicator for wards above the sample median intensity of 5.2, which 10,881 of the 22,354 household-years exceed. Appendix A.5 shows how intensity is distributed across space alongside remittances and production.

Exchange rates. The exchange-rate measure is the rate between the Nepalese rupee and the currency of the migrant's destination country, averaged over the year before the household's interview date. For a household with more than one migrant we take the simple average across its migrants. It varies across households through the destinations they send migrants to, and over time through currency movements and the interview date. We take the series from Google Finance and cross-validate them against OANDA. Two coding conventions follow from aggregating to the household. Migrants who moved within Nepal are assigned a rate of one, because their transfers are denominated in rupees and carry no exchange-rate variation, so a household whose migrants are all internal enters at one. A household with no migrant at all has no destination currency and enters at zero.

3.3 Descriptive Statistics

Table 1 reports means for the main household characteristics and for the outcome blocks we estimate, splitting households by whether they received any remittances during the survey year. About forty percent of household-years receive remittances. Receiving households are smaller, 4.8 against 5.0 members, and far more likely to be headed by a woman, 39 against 13 percent, which is the expected composition effect of adult male out-migration. They are also somewhat more likely to be high-caste and married, and no different in the age of the head. The two groups face the same earthquake intensity on average, an MMI of 5.14 against 5.19, so the remittance split does not stand in for exposure to the earthquake.

Receiving and non-receiving households are statistically indistinguishable in total production, total consumption and total capital inputs. Two margins do differ. Receiving households supply substantially fewer household labor hours, 2,595 against 3,262 a year, and they receive less public assistance, NPR 8,400 against NPR 10,900. Neither raw gap is causal, because households that receive remittances differ in many ways at once, which is what the triple-difference design in Section 4.2 is for. The gaps are nevertheless suggestive of where adjustment happens, in that labor and public transfers are where remittance-receiving households look most different before any of that structure is imposed. The sample is small-scale throughout, with median annual production of NPR 33,000 and three quarters of household-years producing less than NPR 77,000 (Appendix Table A7).

Table 1 — Descriptive statistics: remittance-receiving and non-receiving households

	Non-receiving	Receiving	Full sample	Difference	P-value
Panel A: Household Characteristics					
HH head is high caste	0.10	0.16	0.12	0.06	0.00
HH head is female	0.13	0.39	0.23	0.26	0.00
Age of HH head	48.54	49.06	48.74	0.53	0.10
HH head is married	0.64	0.69	0.66	0.06	0.00
Household Size	5.01	4.80	4.93	-0.21	0.00
Panel B: Earthquake Intensity, Outcomes and Farm Inputs					
Ward earthquake intensity (MMI)	5.19	5.14	5.17	-0.05	0.44
Total Production (in ’000)	80.50	106.03	90.48	25.53	0.53
Total Consumption (in ’000)	359.17	357.38	358.47	-1.79	0.89
Total Capital Inputs (in ’000)	5.76	6.06	5.88	0.30	0.46
Total Labour (hours)	3262	2595	3001	-667	0.00
Total Public Assistance (in ’000)	10.90	8.37	9.91	-2.53	0.00
Household-years	13615	8739	22354		

Notes: One observation per household-year, over the households used in the benchmark production regression. Households are split by whether the household received any remittances in the survey year: 0.39 of household-years receive any, the mean amount conditional on receiving is NPR 186 thousand, and the unconditional mean is NPR 73 thousand. Monetary amounts are in thousands of 2018 Nepalese rupees. Capital inputs, labour hours and public assistance use the same definitions as the corresponding results tables. Difference is the raw gap between the receiving and non-receiving means; the p-value is from a regression of each variable on the remittance indicator with standard errors clustered at the district level.

Appendix Table A6 reports migrants by destination country, together with mean remittances and the mean, standard deviation and coefficient of variation of the exchange-rate measure for each destination. India is the single largest destination and accounts for 41 percent of migrants who went abroad, but its exchange rate is effectively fixed because the Nepalese rupee is pegged to the Indian rupee: its coefficient of variation is 0.00, against 0.02 to 0.19 for the other destinations. Indian migrants also remit far less on average, NPR 44,000 against NPR 167,000 to 187,000 for Malaysia and the Gulf states. The identifying variation therefore comes from the non-Indian destinations, and no single one of them dominates it. Malaysia, Qatar, Saudi Arabia and the United Arab Emirates account for 16, 15, 12 and 6 percent of migrants abroad, with coefficients of variation between 0.05 and 0.14, while the remaining high-income destinations add further dispersion.

4 Empirical Strategy and Results

4.1 Motivating Evidence from Land Cover

Before turning to household remittances, we first ask whether pre-earthquake migration exposure is associated with post-earthquake changes in agricultural land use. The unit of observation is the census ward, indexed by w, which roughly represents a village in Nepal. A simple migration share would be difficult to interpret because migration can respond to local conditions. We therefore use a shift-share measure in the spirit of Khanna et al. (2026) and Yang (2008b), where pre-existing migrant-destination links interact with destination-level changes that shift migrant income and remittance potential.[2] In our setting, wards are more exposed when they had stronger pre-earthquake migrant links to broad destination groups that subsequently experienced faster migration growth elsewhere in Nepal. This measure captures predicted migration and remittance potential rather than realized post-earthquake household choices. We use 2001 and 2011 census data, measured well before the 2015 earthquake, to calculate migration shift-share exposure by weighting each ward's baseline shares across ten migrant-destination groups by subsequent destination-specific migration growth.

2. See Section A.4 for a description of how we implemented the shift-share method.

The growth factors leave out the ward's own district: under the 75-district administrative structure in place during this period, each ward's predicted exposure is shifted by migration growth in the other 74 districts. This makes the exposure less mechanically connected to local agricultural outcomes after the earthquake.

We define HI_w as an indicator for wards with above-median earthquake intensity, measured by Modified Mercalli Intensity, and HSS_w as above-median shift-share migration exposure, distinguishing wards with stronger predetermined exposure to migrant networks whose destination composition was shifting before the earthquake. The outcome is ward cropland area in hectares from the National Land Cover Monitoring System of Nepal (NLCMS) (NLCMS 2022). The estimating equation is a ward-level event-study triple difference: year indicators interacted with HI_w and HSS_w, absorbing ward fixed effects and district-year fixed effects and adding year-interacted core socioeconomic controls and an incomplete-share control. The exercise asks whether earthquake-affected places with greater migration and remittance potential exhibit differential cropland adjustment.

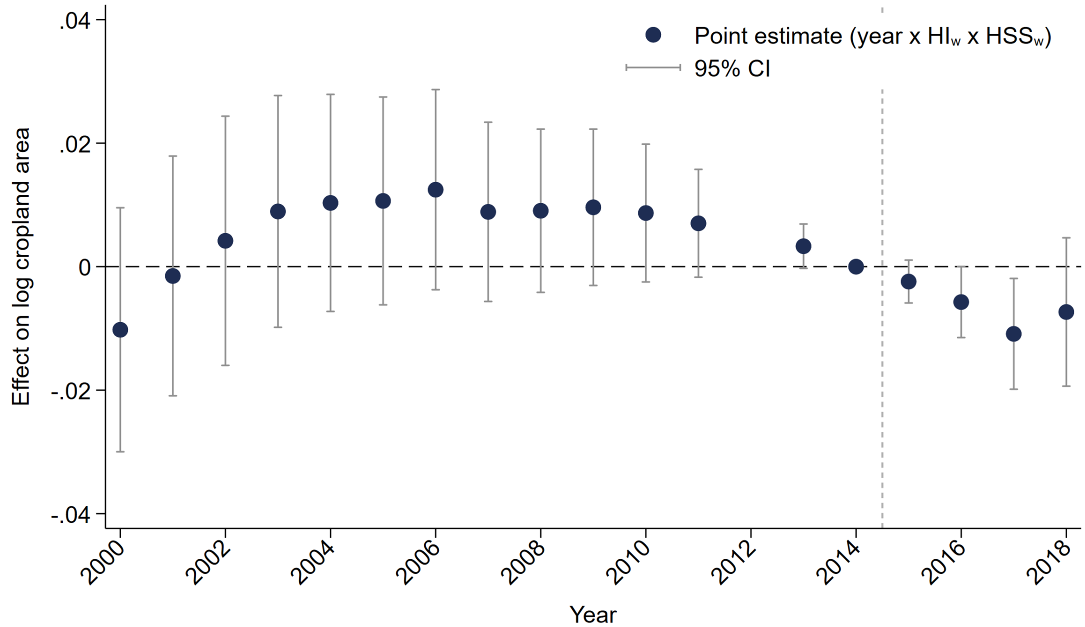

Figure 2 — Cropland event study by earthquake intensity and migration shift-share

Note: The solid dots represent point estimates of the interaction year $\times HI_w \times HSS_w$; the light lines show 95 percent confidence intervals; and the light vertical bar marks the 2015 earthquake. HI_w indicates above-median ward earthquake intensity. HSS_w indicates above-median shift-share migration exposure. The omitted year is 2014, the year immediately before the earthquake. The dependent variable is log ward cropland area, constructed as $\log(C + 0.09)$, where C is cropland hectares from the NLCMS data (NLCMS 2022). A negative point estimate means that cropland area is relatively lower in high-migration-shift-share wards in high-earthquake-intensity locations in that year. The NLCMS data do not report 2012 because several pixels were not scanned due to the failure of a Landsat 7 sensor. The specification includes ward fixed effects, district-year fixed effects, year-interacted core socioeconomic controls, and year interactions with the sum of baseline destination shares. Standard errors are clustered at the district level.

Table 2 — Cropland area and migration shift-share exposure after the earthquake

	Dependent variable: alternative transformations of cropland area C						
	$\log(C+0.09)$	$\text{ihs}(C)$	C (ha)	$\log(C)$ for $C>0$	calibrated $\log(C)$ x=0.1	x=0.5	x=1
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Pooled post-earthquake years 2015–18							
2015–18 x HI_w	0.007	0.006	1.152*	0.008	0.006	0.006	0.006
	(0.006)	(0.006)	(0.681)	(0.006)	(0.006)	(0.006)	(0.006)
2015–18 x HI_w x HSS_w	-0.007*	-0.007**	-0.411	-0.008**	-0.007**	-0.007**	-0.007**
	(0.004)	(0.003)	(0.265)	(0.003)	(0.003)	(0.003)	(0.003)
Panel B. Separate post-earthquake years 2015–18							
2015 x HI_w x HSS_w	-0.002	-0.003*	-0.158	-0.004**	-0.002	-0.002	-0.002
	(0.002)	(0.001)	(0.105)	(0.002)	(0.002)	(0.002)	(0.002)
2016 x HI_w x HSS_w	-0.006*	-0.006**	-0.413*	-0.007***	-0.006**	-0.006**	-0.006**
	(0.003)	(0.003)	(0.213)	(0.003)	(0.003)	(0.003)	(0.003)
2017 x HI_w x HSS_w	-0.011**	-0.010**	-0.650**	-0.011**	-0.009**	-0.009**	-0.009**
	(0.005)	(0.004)	(0.322)	(0.004)	(0.004)	(0.004)	(0.004)
2018 x HI_w x HSS_w	-0.007	-0.009*	-0.421	-0.012*	-0.012*	-0.011*	-0.010*
	(0.006)	(0.006)	(0.475)	(0.006)	(0.006)	(0.006)	(0.006)
N	617922	617922	617922	611598	617922	617922	617922
District	75	75	75	75	75	75	75
Ward and district-year FE	✓	✓	✓	✓	✓	✓	✓
Core controls	✓	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are ward-level NLCMS cropland area, transformed as indicated; C denotes cropland hectares. HI_w indicates above-median ward MMI and HSS_w indicates above-median predicted migration shift-share exposure. The shift-share combines pre-earthquake destination shares with leave-one-district-out destination growth across Nepal's 75 districts. Both panels use the 2001–18 sample and omit 2014. Panel A pools the 2015–18 triple interaction; Panel B displays the separate 2015–18 triple interactions. All specifications include ward fixed effects, district-year fixed effects, year-interacted core socioeconomic controls, and year interactions with baseline destination-share totals. Standard errors are clustered by district. * p<0.10, ** p<0.05, *** p<0.01.

Figure 2 presents the event-study results. The estimates show no discernible differential trend before the earthquake. After the earthquake, cropland area is relatively lower in earthquake-affected wards with higher migration shift-share exposure in every post-earthquake year. Table 2 reports the corresponding core specification for cropland under alternative transformations. Panel A pools the post-earthquake years 2015–18 and reports both the 2015–18 $\times$ HI_w term and its interaction with HSS_w. The pooled triple interaction implies that, after the earthquake, high-intensity wards with higher predetermined migration exposure experienced an additional decline in cropland relative to otherwise comparable wards. Panel B reports the year-specific estimates, which become more negative after the earthquake. Appendix Figure A1 and Appendix Table A8 retain the high-intensity MMI indicator and use shift-share exposure continuously.

Because the shift-share measure combines pre-earthquake destination links with destination growth outside the ward's own district, it is best read as predetermined migration exposure. Given our interest in remittance exposure, however, migration shift-share can only proxy for remittance potential. Baseline migration networks, labor-market access, and local opportunities may still predict agricultural adjustment, and shift-share designs do not automatically remove all such endogeneity concerns (Borusyak, Hull, and Jaravel 2025). The rest of the paper therefore shifts from migration exposure to remittance incomes measured directly in the household surveys and instruments remittance variation with exchange-rate movements. While the land-cover event study motivates the production margin and timing that relate to migration, the IV design below is aimed at identifying the causal effect of remittance responses and, in that regard, isolating the remittance response from overall migration and the other channels through which migration may operate.

4.2 Empirical Strategy

We estimate the effect of remittance responses on post-earthquake outcomes using a triple-difference instrumental-variables design. We use variation across time, across areas with high and low earthquake intensity, and in exchange-rate-induced remittance movements across households. The unit of observation is the household-year. y_{hvt} denotes farm production for household h, which lives in VDC v, in year t, and R_{hvt} denotes the remittances that household receives, summed over its migrants. A household with no migrant carries a remittance of zero. HI_v is an indicator for above-median earthquake intensity. In the pooled specification, we combine the post-earthquake years 2016–2018 and estimate:

$$y_{hvt} = \alpha_v + \tau_t + \boldsymbol{\gamma}\Big(2016\text{–}18 \times HI_v\Big) + \boldsymbol{\beta}\Big(2016\text{–}18 \times HI_v \times R_{hvt}\Big) + \Psi_{hvt} + X'_{hvt}\zeta + \varepsilon_{hvt}, \tag{1}$$

where

$$\Psi_{hvt} = \delta R_{hvt} + \lambda\Big(HI_v \times R_{hvt}\Big) + \eta\Big(2016\text{–}18 \times R_{hvt}\Big).$$

α_v are village development committee (VDC) fixed effects and τ_t are year fixed effects.[3]

The controls X_{hvt} enter at three levels. Household controls are contemporaneous, measured in each round, and cover the ethnicity, age group and marital status of the head. We deliberately exclude household size and its square. A migrant's departure mechanically reduces the size of the household left behind, so household size is partly an outcome of the same migration decision that generates the remittances we instrument; conditioning on it would absorb part of the channel we are trying to measure. Ward controls are fixed at their 2011 census values: ward area, ward poverty, ward population, and the shares of agricultural months and agricultural occupations. Rainfall is a ward-level series averaged over the twelve months before each household's interview date, so it varies across households within the same ward. The main effects of HI_v and 2016–18 are absorbed by the VDC and year fixed effects.[4]

The coefficient $\boldsymbol{\gamma}$ captures the difference-in-differences effect of high earthquake intensity in the post-earthquake period for households with lower remittance inflows, while the coefficient $\boldsymbol{\beta}$ captures how that post-earthquake high-intensity effect changes with remittances. Thus, $\boldsymbol{\beta}$ gives the additional post-earthquake high-intensity effect associated with higher

3. Nepal's administrative hierarchy over this period runs district $\rightarrow$ VDC $\rightarrow$ ward, with 75 districts and roughly nine wards to a VDC. Despite the name, a VDC is a geographic administrative unit rather than a body, and is roughly comparable to a municipality in other countries. We measure earthquake intensity at the ward, one level below the fixed effect. Sampled wards within a VDC almost always fall on the same side of the median intensity, so HI is effectively a VDC-level indicator and we subscript it accordingly.

4. Ψ_{hvt} collects the lower-order remittance terms included in estimation but omitted from the displayed equation.

remittance inflows for otherwise similar households.

To examine how this effect varies across post-earthquake years, we also estimate a year-specific treatment-difference specification. Using 2011 as the omitted base year, we replace the pooled post interaction with explicit post-earthquake year interactions:

$$y_{hvt} = \alpha_v + \tau_t + \sum_{\substack{t=2016 \\ t\neq 2011}}^{2018} \beta_t \Big(\mathbf{1}(t) \times HI_v \times R_{hvt}\Big) + \Psi_{hvt} + X'_{hvt}\zeta + \varepsilon_{hvt}. \tag{2}$$

where

$$\Psi_{hvt} = \delta R_{hvt} + \lambda\Big(HI_v \times R_{hvt}\Big) + \sum_{\substack{t=2016 \\ t\neq 2011}}^{2018} \eta_t\Big(\mathbf{1}(t) \times R_{hvt}\Big) + \sum_{\substack{t=2016 \\ t\neq 2011}}^{2018} \gamma_t\Big(\mathbf{1}(t) \times HI_v\Big).$$

The year-specific tables report only the triple-difference coefficients β_t for 2016, 2017, and 2018, with 2011 as the omitted base year. The pooled coefficient β in (1) is the corresponding aggregate post-earthquake coefficient.

Remittances are endogenous, so we instrument the remittance interaction terms with the corresponding exchange-rate interaction terms. Exchange-rate movements shift the local-currency value of remittances for migrant households while being plausibly orthogonal to household-level production choices in rural Nepal, conditional on fixed effects and controls. The identifying variation therefore comes from exchange-rate-induced shifts in remittances among households differentially affected by the earthquake. Standard errors are clustered at the district level throughout.

4.3 Identification and First Stage

The main identification concern is that remittances are endogenous to household conditions, and selection can run in either direction. More productive households may find it easier to finance a migrant and so receive more remittances, which would bias a raw comparison upward. Migration may equally be a way of compensating for low income or scarce land, and such distress migration biases the comparison downward. Households with stronger networks may also both receive more remittances and recover faster, while remittances may rise in response to local shocks that affect production directly. A simple comparison of remittance-receiving and non-receiving households therefore cannot recover the causal effect of the remittance response to the earthquake, and we have no way to sign its bias. If anything, that ambiguity strengthens the case for instrumenting. We estimate (1) and (2) by two-stage least squares, instrumenting the remittance interaction terms with the corresponding exchange-rate interaction terms, following Yang (2008b).

Table 3 reports the first stage for the pooled treatment-difference remittance term. The dependent variable in all columns is the endogenous interaction 2016–18 $\times HI_v \times R_{hvt}$ under the benchmark transformation. Columns 1 through 4 progressively add the excluded exchange-rate instruments ER_{hvt}, $HI_v \times ER_{hvt}$, 2016–18$\times ER_{hvt}$, and 2016–18$\times HI_v \times ER_{hvt}$. Our preferred specification is the final column because the second stage includes the full set of lower-order remittance terms, and those terms are instrumented with the matching lower-order exchange-rate interactions.

The table therefore reports four excluded instruments even though the main results tables emphasize only the triple-difference coefficient. The lower-order remittance interactions are part of the estimating equation and are instrumented with their own exchange-rate counterparts; they are simply not displayed in the compact second-stage presentation. The main paper keeps the coefficient discussion focused on the post-earthquake triple-difference terms, while the appendix collects the additional variable detail and supporting identification material.

Table 3 — First-stage regressions for the remittance TD interaction

	(1)	(2)	(3)	(4)
ER_{hvt}	0.05***	-0.00	-0.08***	-0.00
	(0.01)	(0.00)	(0.01)	(0.00)
HI_v x ER_{hvt}		0.08***	0.11***	-0.00
		(0.02)	(0.01)	(0.00)
2016-18 x ER_{hvt}			0.10***	-0.00
			(0.01)	(0.00)
2016-18 x HI_v x ER_{hvt}				0.14***
				(0.01)
Endogenous term	R_{hvt}	HI_v x R_{hvt}	2016-18 x R_{hvt}	2016-18 x HI_v x R_{hvt}
SW conditional F	27.71	54.15	219.56	456.42
N	22354	22354	22354	22354
District	71	71	71	71
VDC and year FEs	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Dependent variable is the interaction of the post-earthquake period, high-intensity earthquake exposure, and benchmark-transformed remittances. The excluded exchange-rate (ER_{hvt}) instruments enter sequentially across columns: the ER_{hvt} term, its interaction with HI_v, its interaction with 2016–18, and the full triple interaction. SW conditional F is the Sanderson-Windmeijer conditional F-statistic for the endogenous remittance term named directly above it, computed from the full model in which all four remittance terms are instrumented jointly. The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. *** p<0.01, ** p<0.05, * p<0.10.

Table 3 shows a strong first stage. The Sanderson-Windmeijer conditional F is 27.71

in column (1) and rises to 456.42 in column (4), where the full instrument set is included (Sanderson and Windmeijer 2016). Each endogenous term is therefore separately well predicted by the instrument set, and not merely well predicted on average. This pattern indicates not just that the direct exchange-rate term is relevant on its own, but that the additional high-intensity and post-earthquake interactions are jointly important for predicting the endogenous remittance term used in the preferred specification. It is therefore more accurate to read the progression across columns as evidence of increasing first-stage relevance of the complete excluded-instrument set, rather than as a generic statement about explanatory power.

The more demanding assumption is exclusion restriction: conditional on the fixed effects and controls, exchange-rate movements should affect post-earthquake household outcomes only through remittances. As in Yang (2008b), the identifying idea is that destination-currency fluctuations change the Nepal-rupee value of transfers sent from abroad without being chosen by the recipient household. Relatedly, Khanna et al. (2026) show that exchange-rate shocks can generate plausibly exogenous variation in the migrant income available to origin areas through pre-existing destination exposure. Their estimand is different from ours because they study long-run province-level development using a shift-share design, whereas we use household-destination exchange-rate variation interacted with earthquake exposure; nevertheless, the common logic is that foreign-currency movements can shift the potential pool of migrant income available to the origin economy in ways that are orthogonal to contemporaneous household production decisions. The argument is strengthened by Nepal's macroeconomic setting. Because the Nepalese rupee is pegged to the Indian rupee, domestic households have little scope to influence the bilateral exchange-rate variation embedded in our household-specific instrument; more generally, Shambaugh (2004) shows that small open economies with fixed exchange rates surrender substantial monetary autonomy. In the same spirit, Hull and Imai (2013) emphasize that foreign financial prices can generate plausibly external domestic variation in small open economies. We use that logic narrowly here: not to claim that all foreign shocks are innocuous, but to motivate why the exchange-rate component of remittances is plausibly outside the control of rural Nepali households.

The exclusion restriction would be threatened if exchange-rate movements also affected farm production directly, for example through local output prices, imported input costs, or other local-market channels unrelated to remittance receipts. That concern is limited in our setting. Most households in our sample are smallholders, as Appendix Table A7 shows, so they are not exposed to output-price movements as sellers on any scale. The key identifying variation is also at the household-destination-year level rather than a single market-wide price shock, and any common macroeconomic effects are further absorbed by the time effects and lower-order interaction structure. The remaining identifying requirement is therefore a triple-difference parallel-trends condition: absent the earthquake, the remittance-outcome relationship would have evolved similarly across high- and low-intensity areas, conditional on the included controls and fixed effects. This is the same broad empirical logic used in recent work on Nepal's earthquake that compares high- and low-intensity localities before and after the shock (Pathak and Schündeln 2026), although our estimand is the remittance response rather than the effect of local institutions.

4.4 Main outcome: Household production

Our main outcome is household production income, measured in 2018 Nepalese rupees and constructed by aggregating crop and livestock revenues reported in the survey. Output is a natural object in our setting because the mechanisms of interest operate through input use, labor allocation, crop choice, and production effort that ultimately map into realized production rather than only land productivity. This is also the perspective taken by Burchardi et al. (2019), who study how tenancy incentives affect capital inputs, labor, risk-taking, and output. In our setting, that logic is even more compelling because household production includes both crop and livestock income.

Crop and livestock income are grouped into grains, pulses, vegetables, fruit, meat, milk and eggs, oil, and other agricultural output. Total production is the sum of these categories. Appendix A.1 defines the corresponding outcome blocks and key right-hand-side variables used in the estimating equations.

Production and remittances are highly right-skewed in the data and both contain a non-trivial mass at zero. That matters for estimation because, as emphasized by Chen and Roth (2024), once zeros are present, log-like transformations place different implicit weights on extensive- and intensive-margin changes. Table 4 therefore compares the production results across log, inverse-hyperbolic-sine, extensive-margin, intensive-margin, and calibrated-log specifications. The goal is to separate features of the result that are robust to transformation choice from those that depend on how households moving into or out of positive production are weighted. In the benchmark specification, we use $\log(y + 1)$ for production, remittances, and the exchange-rate instrument so that the same transformation is used consistently across the two stages.

Column (1) uses the $\log(y + 1)$ transformation, which keeps zero-valued observations in the sample through the unit shift and supports a straightforward log-scale interpretation. Column (2) uses the inverse hyperbolic sine transformation, which is defined at zero and behaves like a log for large values, but still mixes extensive and intensive variation in a scale-dependent way. Column (3) reports the extensive margin directly through the indicator for positive production. Column (4) considers the intensive margin by restricting the sample to households with positive production. Columns (5)–(7) implement calibrated-log transformations with alternative values of the extensive-margin parameter x, making the trade-off explicit. Smaller values of x place relatively little weight on changes at zero, while larger values place more weight on the extensive margin.

For the remainder of the paper, we treat column (1), the $\log(y + 1)$ specification, as the benchmark because it keeps zero-valued outcomes in the sample and lets us interpret the estimates on a common log scale across the outcome, remittances, and the instrument. At the same time, Chen and Roth (2024) show that zeros make transformation choice substantive rather than innocuous. For that reason, we continue to report $\text{ihs}(y)$ and calibrated-log specifications as explicit robustness checks. In our data, however, the benchmark conclusions are not driven by that choice: the negative production effect is similar under $\log(y+1)$, $\text{ihs}(y)$, and calibrated-log transformations, especially for total production.

Our preferred estimate therefore comes from column (1). The coefficient is about -0.41, while the comparable calibrated-log estimates range from about -0.42 to -0.45, which suggests that the result is not an artifact of one particular way of handling zeros. Because both production and remittances enter as $\log(y + 1)$, the estimate is most naturally read on a log scale: higher remittance inflows are associated with lower total production, with an elasticity close to -0.41 away from zero. Because the benchmark transformation keeps zero-producing households in the sample, that estimate still combines two channels: fewer households remaining in positive production, and lower production among households that continue to produce. The negative extensive-margin result in column (3) and the weaker intensive-margin result in column (4) suggest that the first channel is especially important. In that sense, the remittance response to the earthquake appears to operate more strongly through continued participation in production than through lower output among households that remain producers. Appendix Tables A3 and A4 further show that this total-outcome pattern is similar when remittances and exchange rates enter in levels and when Kathmandu Valley districts are excluded.

Table 4 — Total production under alternative outcome transformations

	Dep var: alternative transformations of total production (P)						
	$\log(P+1)$	ihs(P)	$1(P>0)$	log(P) for $P>0$	calibrated log(P) x=0.1	x=0.5	x=1
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Pooled post-earthquake years 2016–18							
2016-18 x $\mathrm{HI_v}$	0.24	0.28	0.05	-0.19	0.25	0.27	0.29
	(0.91)	(0.96)	(0.08)	(0.24)	(0.92)	(0.95)	(0.99)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.41***	-0.44***	-0.04***	-0.05	-0.42***	-0.43***	-0.45***
	(0.14)	(0.15)	(0.01)	(0.06)	(0.14)	(0.14)	(0.15)
Panel B. Separate post-earthquake years 2016–18							
2016 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.41***	-0.44***	-0.04***	-0.06	-0.42***	-0.43***	-0.45***
	(0.15)	(0.15)	(0.01)	(0.06)	(0.15)	(0.15)	(0.16)
2017 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.44***	-0.47***	-0.04***	-0.02	-0.45***	-0.46***	-0.49***
	(0.15)	(0.15)	(0.01)	(0.06)	(0.15)	(0.15)	(0.16)
2018 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.36**	-0.38**	-0.03**	-0.07	-0.36**	-0.38**	-0.39**
	(0.14)	(0.15)	(0.01)	(0.06)	(0.14)	(0.15)	(0.16)
N	22354	22354	22354	18064	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓

Notes: Dependent variable is total production, transformed as indicated in the column heading. Columns (5)–(7) use calibrated-log production with alternative extensive-margin values x. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 5 then decomposes the benchmark production outcome into its constituent categories using the same $\log(y+1)$ specification. This component decomposition shows whether the total production response reflects a broad-based decline across household production activities or changes concentrated in a narrower set of crop and livestock categories.

Table 5 suggests that the aggregate decline is not driven by a single idiosyncratic category. Most categories move in the negative direction, but the clearest estimated declines appear in grains and meat. Grains are the largest component of household production in this setting and depend on purchased seed and fertilizer at planting, while meat requires sustained outlays on animals and feed. Both are therefore categories for which continued production involves ongoing commitments of cash, labor, or working capital. The component pattern is consistent with a post-earthquake reduction in relatively input-intensive production activities. The broadly negative but less precise estimates for the other components suggest that the total effect is diffuse rather than confined to one narrow crop or livestock margin.

Table 5 — Production components and total

	Dependent variable: log(production variables + 1)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	grains	pulses	veges	fruits	meat	milk egg	oil	other	total
	Panel A. Pooled post-earthquake years 2016–18								
2016-18 x $\mathrm{HI_v}$	0.46	-0.40	0.73	0.64	0.81***	-0.62	-0.05	0.42	0.24
	(0.84)	(0.71)	(0.91)	(0.74)	(0.26)	(0.58)	(0.45)	(0.87)	(0.91)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}} + 1)$ x $\mathrm{HI_v}$	-0.33**	-0.01	-0.17	-0.14	-0.14***	-0.07	-0.03	-0.20*	-0.41***
	(0.13)	(0.09)	(0.11)	(0.12)	(0.04)	(0.10)	(0.07)	(0.12)	(0.14)
	Panel B. Separate post-earthquake years 2016–18								
2016 x $\log(\widehat{R_{\mathrm{hvt}}} + 1)$ x $\mathrm{HI_v}$	-0.31**	0.06	-0.12	-0.16	-0.12**	-0.13	-0.05	-0.23*	-0.41***
	(0.14)	(0.11)	(0.12)	(0.12)	(0.05)	(0.12)	(0.09)	(0.13)	(0.15)
2017 x $\log(\widehat{R_{\mathrm{hvt}}} + 1)$ x $\mathrm{HI_v}$	-0.36**	-0.09	-0.25**	-0.14	-0.14***	-0.07	-0.01	-0.16	-0.44***
	(0.14)	(0.12)	(0.13)	(0.12)	(0.05)	(0.11)	(0.08)	(0.13)	(0.15)
2018 x $\log(\widehat{R_{\mathrm{hvt}}} + 1)$ x $\mathrm{HI_v}$	-0.29**	-0.00	-0.12	-0.10	-0.14***	-0.00	-0.04	-0.20	-0.36**
	(0.14)	(0.10)	(0.12)	(0.12)	(0.05)	(0.13)	(0.08)	(0.13)	(0.14)
N	22354	22354	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are the natural logarithm of one plus each production component and the natural logarithm of one plus total production. The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

4.5 Mechanisms

4.5.1 Consumption Smoothing

If higher remittance inflows after the earthquake were associated with a tighter household budget constraint, we would expect consumption to fall alongside production. That is not what the data suggest. Total consumption does decline, but by far less than production: -0.07 against -0.41 on the same log scale, with the decline appearing in grains, vegetables, fruit and the residual category. A contraction of that size alongside a production fall six times larger is more consistent with consumption smoothing than with a broad contraction in both. The evidence points to consumption that is close to maintained while production falls sharply.

That interpretation is consistent with standard insurance logic in the remittances literature. Yang and Choi (2007) show that remittances can respond to adverse shocks in ways that stabilize recipient-household consumption, and Combes and Ebeke (2011) emphasize the broader consumption-smoothing role of remittances in shock-prone environments. In Nepal specifically, Mishra, Kondratjeva, and Shively (2022) find that remittances are associated with higher food expenditures in normal times. Our results fit that broader literature in a post-disaster setting: consumption moves comparatively little, while the adjustment appears in agricultural production.

Table 6 — Consumption components and total

	Dependent variable: log(consumption variables + 1)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	grains	pulses	veges	fruits	meat	milk egg	oil	other	total
	Panel A. Pooled post-earthquake years 2016–18								
2016-18 x $\mathrm{HI_v}$	0.12	0.26	0.05	1.38**	0.26	-0.92	0.07	0.02	0.06
	(0.12)	(0.32)	(0.16)	(0.58)	(0.57)	(0.85)	(0.17)	(0.34)	(0.09)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.07***	-0.14	-0.06*	-0.26*	0.08	-0.20	-0.05	-0.21**	-0.07***
	(0.02)	(0.09)	(0.03)	(0.15)	(0.17)	(0.17)	(0.03)	(0.09)	(0.02)
	Panel B. Separate post-earthquake years 2016–18								
2016 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.06***	-0.17*	-0.05	-0.26	0.02	-0.14	-0.05	-0.32***	-0.07***
	(0.02)	(0.10)	(0.03)	(0.19)	(0.18)	(0.20)	(0.03)	(0.10)	(0.02)
2017 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.06***	-0.10	-0.04	-0.25*	0.01	-0.27	-0.04	-0.20*	-0.06***
	(0.02)	(0.09)	(0.03)	(0.13)	(0.18)	(0.19)	(0.03)	(0.11)	(0.02)
2018 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.07***	-0.15	-0.07**	-0.29	0.22	-0.19	-0.04	-0.08	-0.07***
	(0.02)	(0.09)	(0.03)	(0.18)	(0.20)	(0.17)	(0.04)	(0.12)	(0.02)
N	22354	22354	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are the natural logarithm of one plus each consumption component and the natural logarithm of one plus total consumption. The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

4.5.2 Capital Inputs

The capital-input results speak directly to the production-technology channel. If remittances after the earthquake make it easier for households to cope without maintaining the same level of agricultural activity, the first place we should see that is in the purchased inputs that support production: seeds, fertilizer, irrigation expenses, and equipment rental. These are the inputs through which households convert intended production into realized output, so they provide direct evidence on whether the main production result operates through changes in production scale rather than through consumption smoothing alone.

Table 7 asks whether households with higher remittance inflows in high-intensity areas after the earthquake reduced capital spending on the farm. The estimates point most clearly to lower spending on seeds and on the capital-input total, while the other components also tend to be negative but less precisely estimated. That pattern is consistent with a broad but uneven decline in production intensity. Seeds are especially informative because they are among the earliest and most direct cash commitments households make when maintaining crop production.

In our setting, capital spending is an especially important margin because production requires upfront expenditures. Negative capital-input coefficients therefore help explain why total production falls: households are not simply harvesting less from a fixed technology, but are using less of the purchased and cash-intensive inputs that sustain production.

Table 7 — Capital input components and total

	Dependent variable: log(capital input variables + 1)				
	(1)	(2)	(3)	(4)	(5)
	seeds	fertilizer	irrigation	rented equipment	total
	Panel A. Pooled post-earthquake years 2016–18				
2016-18 x $\mathrm{HI_v}$	0.02	0.18	-0.16	-1.51**	-0.34
	(0.61)	(0.53)	(0.58)	(0.61)	(0.71)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.19*	-0.12	-0.05	-0.00	-0.22*
	(0.10)	(0.10)	(0.09)	(0.11)	(0.13)
	Panel B. Separate post-earthquake years 2016–18				
2016 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.16	-0.19*	-0.03	-0.07	-0.32**
	(0.12)	(0.11)	(0.08)	(0.14)	(0.14)
2017 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.16	-0.14	-0.08	0.10	-0.17
	(0.11)	(0.09)	(0.11)	(0.13)	(0.13)
2018 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.24*	0.01	-0.06	-0.03	-0.16
	(0.13)	(0.15)	(0.11)	(0.13)	(0.14)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓

Notes: Dependent variables are the natural logarithm of one plus each capital-input component and the natural logarithm of one plus total capital input. The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

4.5.3 Labor Inputs

Labor inputs provide a second mechanism. We ask whether higher remittance inflows after the earthquake change how households allocate labor across agricultural activities. If the main production effect reflects a genuine reallocation away from farming, we should observe weaker labor inputs as well, either because households withdraw their own time from agriculture, because they hire less outside labor, or both.

Table 8 focuses on hired labor spending and the imputed value of self-supplied agricultural labor. This distinction matters. A decline in hired labor suggests that households are cutting purchased labor inputs as part of a broader reduction in production intensity. A decline in self-agricultural labor value points more directly to a reduction in household labor devoted to agriculture.

In our estimates, the decline is concentrated in the household's own labor: the value of self-supplied agricultural labor falls significantly, as does total labor input, while hired labor spending shows no significant change. Consistent with this, the labor-allocation results in Appendix Table A5 show that household members' wage agricultural labor also declines. That pattern strengthens the interpretation that the production effect operates through lower own agricultural effort alongside the lower purchased inputs documented in Table 7.

Table 8 — Labor input components and total

	Dep var: log(labor input vars + 1)		
	(1) hired labor	(2) self agriculture	(3) total
	Panel A. Pooled post-earthquake years 2016–18		
2016-18 x HI_v	-0.74	-0.75	-0.82
	(0.76)	(0.69)	(0.74)
2016-18 x $\log(\widehat{R_{hvt}+1})$ x HI_v	0.09	-0.25*	-0.26**
	(0.11)	(0.13)	(0.13)
	Panel B. Separate post-earthquake years 2016–18		
2016 x $\log(\widehat{R_{hvt}+1})$ x HI_v	0.08	-0.22	-0.22
	(0.11)	(0.14)	(0.14)
2017 x $\log(\widehat{R_{hvt}+1})$ x HI_v	0.12	-0.33**	-0.34**
	(0.14)	(0.14)	(0.14)
2018 x $\log(\widehat{R_{hvt}+1})$ x HI_v	0.09	-0.20	-0.22
	(0.13)	(0.13)	(0.13)
N	22354	22354	22354
District	71	71	71
VDC and year FEs	✓	✓	✓
Ψ_{hvt}	✓	✓	✓
Controls	✓	✓	✓

Notes: Dependent variables are log(1 + hired labor spending), log(1 + self-agricultural labor value), and log(1 + their total). The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

4.5.4 Public Assistance

Next, we ask whether remittance responses are associated with public-assistance receipts after the shock. This block reports earthquake aid first, followed by old-age/widow, disability/ethnic, maternal/child, conflict/martyr, other aid, and their total.

Public assistance provides a complementary coping margin to private remittances. It may partly offset household losses after the earthquake, but it is institutionally distinct from private transfers and household production choices.

Table 9 — Public assistance components and total

	Dependent variable: log(public assistance + 1)						
	(1) earthquake aid	(2) old age widow	(3) disability ethnic	(4) maternal child	(5) conflict martyr	(6) other aid	(7) total
	Panel A. Pooled post-earthquake years 2016–18						
2016-18 x $\mathrm{HI_v}$	2.11**	0.06	-0.03	-0.18	-0.01	-0.01*	1.73**
	(0.87)	(0.30)	(0.09)	(0.11)	(0.01)	(0.01)	(0.76)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.03	-0.12	-0.00	0.05	0.00	0.00	-0.09
	(0.03)	(0.11)	(0.02)	(0.03)	(0.00)	(0.00)	(0.11)
	Panel B. Separate post-earthquake years 2016–18						
2016 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.08*	-0.08	0.00	0.07*	0.01*	0.00	-0.11
	(0.04)	(0.11)	(0.02)	(0.04)	(0.00)	(0.00)	(0.12)
2017 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	0.01	-0.19*	-0.02	0.02	0.00	-0.01	-0.15
	(0.05)	(0.11)	(0.02)	(0.03)	(0.00)	(0.01)	(0.13)
2018 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	0.02	-0.19	0.02	0.06	0.00	0.01	-0.08
	(0.05)	(0.13)	(0.02)	(0.04)	(0.00)	(0.01)	(0.13)
N	22354	22354	22354	22354	22354	22354	22354
District	71	71	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are log(1 + public-assistance component) and log(1 + total public assistance). The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

4.5.5 Mediation Analysis

Our IV strategy is aimed at identifying the total effect of the remittance response to the earthquake on production income. Following the logic of Imai, Keele, and Tingley (2010), Table 10 shows a one-mediator-at-a-time decomposition of the triple-difference IV effect through total consumption, total capital inputs, total labor inputs, and total public assistance. This exercise allows us to examine more precisely which factors mediate the effect of post-disaster remittances on production. A causal interpretation of the mediator decomposition requires a sequential-ignorability assumption (Imai, Keele, and Tingley 2010). In our context, we should be able to argue that these mediators are as good as random with respect to the potential production outcome after conditioning on treatment, controls, and fixed effects. We are aware that this requirement is quite demanding here because consumption, capital use, labor allocation, and public assistance are all post-earthquake household choices or transfers that are likely to move with unobserved recovery needs, local conditions, expectations, and other coping responses. Therefore, we do not claim point identification of a fully causal mediation effect under these weak assumptions. Rather, we use this exercise to summarize how much of the total production effect is statistically associated with each mediator. The results show that several post-earthquake adjustment margins move in ways that are consistent with the main production result using the same IV variation as the main design rather than establishing a fully causal channel decomposition.

The results show that mediation through total consumption accounts for about a fifth of the total effect, well below the input margins, which reinforces the earlier conclusion that giving up on consumption is not the main margin explaining lower production. The indirect effects through capital and labor are considerably larger, accounting for roughly two-fifths and just under half of the total production effect respectively. Public assistance does not appear to mediate any of it. Taken together, the mediation evidence reinforces earlier findings, namely, the decline in production is linked much more closely to lower input use and reduced agricultural labor than to consumption or public assistance. We note that the mediated shares are estimated one mediator at a time and are not constrained to sum to the total effect, so they should be read as the relative importance of each margin rather than as an exhaustive decomposition.

Table 10 — IV mediation of total production through transformed totals

	Dependent variable: log(total production + 1)				
	Reference	Total consumption	Total capital	Total labor	Total public assistance
Total effect (TD)	-0.41***	-0.41***	-0.41***	-0.41***	-0.41***
	(0.14)	(0.14)	(0.14)	(0.14)	(0.14)
Indirect effect		-0.08***	-0.15**	-0.19***	-0.00
		(0.02)	(0.11)	(0.12)	(0.00)
Direct effect		-0.33***	-0.26***	-0.23***	-0.41***
		(0.15)	(0.07)	(0.08)	(0.14)
Proportion mediated		19.55***	37.56**	45.07***	0.05
		(8.16)	(15.20)	(18.44)	(0.23)
N	22354	22354	22354	22354	22354

Notes: The outcome is log(total production + 1). Column 1 reports the baseline total effect from the benchmark production specification. Columns 2–5 report one-mediator-at-a-time decompositions through benchmark-transformed totals for consumption, capital, labor, and public assistance. Mediator totals follow the definitions used in the main mechanism tables. Within each mediator, the total, indirect, and direct effects are estimated on a common mediator-specific sample. The remittance TD interaction and its lower-order terms are instrumented with the corresponding exchange-rate interactions, the exchange rate entering in levels. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Household size and its square are excluded as bad controls. Standard errors are bootstrap standard deviations across 50 district-clustered replications; replications in which a parameter could not be estimated are excluded, and the number completed is reported in the log. Stars are based on two-sided sign-based clustered-bootstrap p-values. Proportion mediated is reported in percent and can be unstable when the total effect is close to zero in bootstrap draws.

5 Conclusion

This paper studies how the remittance response to a large shock affects household production. Using exchange-rate-driven variation in remittances within a triple-difference IV design, we show that higher remittance inflows after the Nepal earthquake reduced household production in the short run. Read on its own, that result could be easy to misinterpret. The broader pattern in the data is more specific. Total consumption falls by far less than production, while capital inputs and agricultural labor decline, and the component results point most clearly to categories that tend to require continued cash, labor, or working-capital commitments. The mediation exercise, while more demanding in its identifying assumptions, points in the same direction: the production effect lines up more closely with capital and labor adjustment than with lower consumption.

Taken together, the evidence suggests that remittances may help households cope with the shock through alternative margins, but do not appear to support the maintenance of pre-earthquake production levels, capital, or input use. Instead, higher remittance inflows seem to ease short-run budget constraints, making it more feasible for households to scale back agricultural activity when sustaining production is no longer the primary margin of adjustment. This interpretation is consistent with largely maintained consumption alongside much larger declines in input use and agricultural labour. At the same time, the results do not isolate a single mechanism. Rather, they point to a broader pattern of post-disaster adjustment: remittances relax the need to maintain farm production at pre-shock levels.

These findings suggest that private remittances are not a substitute for policies aimed at maintaining production at the pre-disaster levels. If anything, they may enable households to reduce the need to sustain production. Supporting recovery after a major shock therefore requires targeted interventions on the margins where production declines, particularly inputs, labour, and short-run working capital. In addition to policies that help households smooth welfare, there remains a clear role for policies that support the maintenance of production at pre-disaster levels.

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A Appendix

A.1 Variable Definitions and Transformations

This appendix defines the main outcome blocks and reports robustness checks for the functional-form choices used in the text. The benchmark specification uses $\log(y + 1)$, which keeps zero-valued observations in the estimating sample and puts production, remittances, and exchange rates on a common log scale. Appendix Tables A1 and A2 report the corresponding total-outcome estimates under calibrated-log and $\text{ihs}(y)$ transformations. Appendix Tables A3 and A4 show that the main total-outcome patterns are similar when remittances and exchange rates enter in levels and when Kathmandu Valley districts are excluded.

Production. Household production is crop and livestock income measured in 2018 Nepalese rupees. The component table groups production into broad agricultural categories such as grains, pulses, vegetables, fruit, livestock products, oil, and other output. Total production is the sum of these components.

Consumption. Consumption is grouped into comparable food categories, allowing the mechanism tables to distinguish production responses from changes in household food consumption. Total consumption is the sum of these categories.

Capital inputs. Capital inputs capture purchased or cash-intensive agricultural inputs, including seeds, fertilizer, irrigation, and rented equipment. Total capital input is the sum of these expenditures.

Labor inputs. Labor inputs capture both hired agricultural labor and the value of household self-employment in agriculture. Appendix Table A5 extends this margin by separating agricultural and non-agricultural labor allocation.

Public assistance. Public assistance captures government and program transfers received after the earthquake, including earthquake relief and other social assistance. Total public assistance is the sum of these transfer categories.

Main right-hand-side variables. The treatment-intensity variable HI_v is the indicator for high earthquake intensity. The pooled post period is the indicator for years 2016–18. Household remittance exposure enters as the transformed remittance measure R_{hvt}, and the corresponding exchange-rate measure enters as the instrument. In the benchmark specification, the same transformation family is applied to the outcome, remittance, and exchange-rate variables, so the second-stage coefficient and the first-stage instrument operate on a common functional-form scale. The control set combines household demographics, rainfall, ward characteristics, and baseline agricultural conditions.

Table A1 — Appendix Table A1. Totals under calibrated log(y)

	Dependent variable: calibrated log(total outcome)			
	(1)	(2)	(3)	(4)
	Production	Consumption	Capital	Labor
	Panel A. Pooled post-earthquake years 2016–18			
2016-18 x $\mathrm{HI_v}$	-0.15	0.00	-0.57	-0.98
	(0.81)	(0.10)	(0.64)	(0.63)
2016-18 x cal. $\widehat{\widetilde{\log}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.23***	-0.04***	-0.12	-0.19**
	(0.07)	(0.01)	(0.08)	(0.08)
	Panel B. Separate post-earthquake years 2016–18			
2016 x cal. $\widehat{\widetilde{\log}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.24***	-0.05***	-0.22**	-0.16*
	(0.09)	(0.01)	(0.09)	(0.09)
2017 x cal. $\widehat{\widetilde{\log}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.27***	-0.04**	-0.05	-0.26***
	(0.08)	(0.02)	(0.08)	(0.09)
2018 x cal. $\widehat{\widetilde{\log}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.16*	-0.04**	-0.06	-0.15*
	(0.08)	(0.02)	(0.09)	(0.08)
N	22354	22354	22354	22354
District	71	71	71	71
VDC and year FEs	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Each column reports the total outcome from the corresponding block using the calibrated-log transformation. Calibrated-log specifications assign zero outcomes a value of -0.1, following the extensive-margin calibration logic discussed in Chen and Roth. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A2 — Appendix Table A2. Totals under ihs(y)

	Dependent variable: ihs(total outcome)			
	(1)	(2)	(3)	(4)
	Production	Consumption	Capital	Labor
	Panel A. Pooled post-earthquake years 2016–18			
2016-18 x $\mathrm{HI_v}$	-0.17	0.02	-0.63	-1.05
	(0.85)	(0.10)	(0.69)	(0.67)
2016-18 x $\widehat{\mathrm{ihs}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.18***	-0.04***	-0.09	-0.15**
	(0.07)	(0.01)	(0.07)	(0.07)
	Panel B. Separate post-earthquake years 2016–18			
2016 x $\widehat{\mathrm{ihs}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.19**	-0.05***	-0.17**	-0.13
	(0.08)	(0.01)	(0.08)	(0.09)
2017 x $\widehat{\mathrm{ihs}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.22***	-0.04***	-0.02	-0.20**
	(0.07)	(0.01)	(0.07)	(0.08)
2018 x $\widehat{\mathrm{ihs}(R_{\mathrm{hvt}})}$ x $\mathrm{HI_v}$	-0.12	-0.03***	-0.03	-0.12*
	(0.07)	(0.01)	(0.07)	(0.07)
N	22354	22354	22354	22354
District	71	71	71	71
VDC and year FEs	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Each column reports the total outcome from the corresponding block using the inverse hyperbolic sine transformation. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A3 — Benchmark totals using remittances and exchange rates in levels

	Dependent variable: log(total outcome + 1)			
	(1)	(2)	(3)	(4)
	Production	Consumption	Capital	Labor
	Panel A. Pooled post-earthquake years 2016–18			
2016-18 x $\mathrm{HI_v}$	0.08	-0.09	-0.52	-0.98
	(1.34)	(0.13)	(0.95)	(1.05)
2016-18 x $\widehat{R}_{\mathrm{hvt}}$ x $\mathrm{HI_v}$	-0.01**	-0.00**	-0.00	-0.00*
	(0.00)	(0.00)	(0.00)	(0.00)
	Panel B. Separate post-earthquake years 2016–18			
2016 x $\widehat{R}_{\mathrm{hvt}}$ x $\mathrm{HI_v}$	-0.01**	-0.00**	-0.01*	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)
2017 x $\widehat{R}_{\mathrm{hvt}}$ x $\mathrm{HI_v}$	-0.01**	-0.00*	-0.00	-0.01**
	(0.00)	(0.00)	(0.00)	(0.00)
2018 x $\widehat{R}_{\mathrm{hvt}}$ x $\mathrm{HI_v}$	-0.01*	-0.00*	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)
N	22354	22354	22354	22354
District	71	71	71	71
VDC and year FEs	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Each column reports the natural logarithm of one plus the total outcome from the corresponding block, but the endogenous remittance term and the excluded exchange-rate instrument enter in levels. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A4 — Benchmark totals excluding Kathmandu Valley districts

	Dependent variable: log(total outcome + 1)			
	(1)	(2)	(3)	(4)
	Production	Consumption	Capital	Labor
Panel A. Pooled post-earthquake years 2016–18				
2016-18 x $\mathrm{HI_v}$	-0.22	-0.02	-0.65	-1.02
	(0.81)	(0.10)	(0.65)	(0.63)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.21**	-0.03**	-0.10	-0.18**
	(0.08)	(0.01)	(0.08)	(0.08)
Panel B. Separate post-earthquake years 2016–18				
2016 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.21**	-0.04**	-0.19**	-0.15
	(0.09)	(0.01)	(0.09)	(0.10)
2017 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.25***	-0.03*	-0.03	-0.24**
	(0.08)	(0.02)	(0.08)	(0.10)
2018 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.14	-0.02	-0.04	-0.15*
	(0.08)	(0.02)	(0.08)	(0.08)
N	21383	21383	21383	21383
District	68	68	68	68
VDC and year FEs	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Each column reports the natural logarithm of one plus the total outcome from the corresponding block after excluding Kathmandu, Lalitpur, and Bhaktapur. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A5 — Labor allocation across agriculture and non-agriculture

	Dependent variable: log(labor-allocation variables + 1)				
	(1) wage agri.	(2) wage non-agri.	(3) self agri.	(4) self non-agri.	(5) total
	Panel A. Pooled post-earthquake years 2016–18				
2016-18 x $\mathrm{HI_v}$	0.31	0.66	-0.62	0.76*	-0.33
	(0.34)	(0.53)	(0.42)	(0.45)	(0.33)
2016-18 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.17***	-0.14	-0.15*	0.04	0.13
	(0.05)	(0.16)	(0.08)	(0.16)	(0.15)
	Panel B. Separate post-earthquake years 2016–18				
2016 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.19***	-0.17	-0.14	0.04	0.11
	(0.05)	(0.17)	(0.09)	(0.16)	(0.16)
2017 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.14**	-0.16	-0.18**	0.06	0.14
	(0.06)	(0.17)	(0.09)	(0.16)	(0.16)
2018 x $\log(\widehat{R_{\mathrm{hvt}}}+1)$ x $\mathrm{HI_v}$	-0.18***	-0.11	-0.14	0.02	0.14
	(0.05)	(0.17)	(0.08)	(0.17)	(0.15)
N	22354	22354	22354	22354	22354
District	71	71	71	71	71
VDC and year FEs	✓	✓	✓	✓	✓
Ψ_{hvt}	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓

Notes: Dependent variables are the natural logarithm of one plus each labor-allocation component and the natural logarithm of one plus total labor allocation. The benchmark specification uses the natural logarithm of one plus the outcome, remittances, and the exchange-rate instrument. All specifications report two-stage least squares estimates with high-dimensional fixed effects. The endogenous remittance terms are instrumented with the corresponding exchange-rate interactions. In Panel B, 2011 is the omitted base year. Controls include ethnicity, age group, marital status, average annual rainfall, ward area, ward poverty, ward population, and baseline agriculture controls. Household size and its square are excluded as bad controls: a migrant's departure mechanically reduces household size, so conditioning on it absorbs part of the remittance channel. The estimation sample is the full set of household-years. The exchange rate is the simple mean over the household's migrant rows; a household with only internal migrants carries one, and a household with no migrant carries zero. Standard errors are clustered at the district level. Standard errors are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

A.2 Migrant Destinations and Exchange-Rate Variation

Table A6 documents where the exchange-rate variation in the instrument comes from. Each row is a destination country, with the currency used to construct the rate, the number of migrant person-years in the estimation sample, mean remittances, and the mean, standard deviation and coefficient of variation of the exchange-rate measure over the sample period. Because currencies are quoted in different units, the coefficient of variation is the comparable measure of movement across rows. India is reported for completeness even though the peg leaves it with no usable variation, and Bhutan is reported at the Indian rupee rate because the ngultrum is pegged at par with the Indian rupee. Migrants whose destination is recorded only as "other" cannot be assigned a currency and therefore have no exchange rate, so they fall outside the estimation sample; this costs 615 migrant person-years, 2.9 percent of all migrants in the panel.

Table A6 — Migrant destinations, remittances, and exchange-rate variation

Destination	Currency	Obs.	Share (%)	Mean remit. (in '000)	Mean ER	SD ER	CV ER
India	INR	3997	41.1	44.4	1.58	0.00	0.00
Malaysia	MYR	1599	16.4	166.9	24.45	1.25	0.05
Qatar	QAR	1486	15.3	181.1	26.41	3.68	0.14
Saudi Arabia	SAR	1194	12.3	184.8	25.92	3.47	0.13
United Arab Emirates	AED	562	5.8	186.7	26.36	3.54	0.13
Japan	JPY	211	2.2	327.0	0.92	0.05	0.05
United States	USD	184	1.9	197.1	84.26	14.97	0.18
United Kingdom	GBP	135	1.4	184.3	118.21	9.56	0.08
Australia	AUD	122	1.3	177.5	70.92	6.98	0.10
South Korea	KRW	113	1.2	542.8	0.09	0.01	0.12
Hong Kong	HKD	79	0.8	92.3	10.89	1.95	0.18
Israel	ILS	24	0.2	338.8	21.76	3.88	0.18
China	CNY	12	0.1	48.9	11.26	2.12	0.19
Bangladesh	BDT	6	0.1	0.0	1.06	0.03	0.02
Bhutan	INR	1	0.0	0.0	1.58	–	–
Total		9725	100.0				

Notes: Migrant person-years with an international destination, restricted to the households in the estimation sample, by destination country. A household with migrants in two countries contributes to two rows, so the unit here is the migrant rather than the household. ER is the exchange-rate measure attached to the migrant's own destination (Nepalese rupees per unit of destination currency, average annual rate prior to interview); CV is the coefficient of variation (SD/mean), which makes exchange-rate movements comparable across currencies with different units. Bhutan is reported at the Indian rupee rate because the ngultrum is pegged at par with the Indian rupee. Because the Nepalese rupee is likewise pegged to the Indian rupee, India shows almost no exchange-rate variation, so the identifying variation in the instrument comes from the other destinations despite India accounting for the largest share of migrants.

Table A7 reports the distribution of total household production across the household-years in the estimation sample. The concentration of the distribution at low values is what we mean in describing the sample as smallholder: three quarters of household-years produce less than NPR 77,000 of crop and livestock output in a year.

Table A7 — Distribution of total household production

Percentile	25th	50th	75th	90th	95th	99th
Total production (in '000)	8.5	33.3	76.9	149.6	229.2	599.0

Notes: Percentiles of total household production (crop and livestock, in thousands of 2018 Nepalese rupees) in the estimation sample.

A.3 Mediation Framework

This appendix section formalizes the mediation exercise reported in Table 10. The setup follows the decomposition logic in Imai, Keele, and Tingley (2010), but is adapted to our triple-difference IV specification and should therefore be read as a structured IV mediation analysis rather than a fully nonparametric mediation design.

Let Y_{hvt} denote transformed total household production, and let M^k_{hvt} denote one candidate mediator, where $k \in \{\text{consumption total, capital total, labor total, public-assistance total}\}$. For each mediator, we estimate three equations. First, the total effect equation is

$$Y_{hvt} = \alpha_v + \tau_t + \beta_1\Big(Post_t \times HI_v \times R_{hvt}\Big) + \Psi_{hvt} + X'_{hvt}\zeta_1 + \varepsilon_{1,hvt}, \tag{A.1}$$

where Ψ_{hvt} collects the lower-order remittance terms and their interactions already included in the main specification.

Second, for each mediator M^k_{hvt} we estimate

$$M^k_{hvt} = \alpha_v + \tau_t + \beta_{2k}\Big(Post_t \times HI_v \times R_{hvt}\Big) + \Psi_{hvt} + X'_{hvt}\zeta_{2k} + \varepsilon_{2k,hvt}. \tag{A.2}$$

Third, we include the mediator directly in the production equation:

$$Y_{hvt} = \alpha_v + \tau_t + \beta_{3k}\Big(Post_t \times HI_v \times R_{hvt}\Big) + \xi_k M^k_{hvt} + \Psi_{hvt} + X'_{hvt}\zeta_{3k} + \varepsilon_{3k,hvt}. \tag{A.3}$$

In all three equations, the remittance interaction terms are instrumented with the corresponding exchange-rate interaction terms, exactly as in the main IV specification. For mediator k, the reported decomposition is

$$\text{TE}_k = \beta_1, \qquad \text{NIE}_k = \beta_{2k}\xi_k, \qquad \text{NDE}_k = \beta_{3k}, \qquad \text{PM}_k = 100 \times \frac{\text{NIE}_k}{\text{TE}_k}.$$

Standard errors are obtained by district-clustered bootstrap.

The interpretation of these quantities requires care. In the language of Imai, Keele, and Tingley (2010), a causal mediation interpretation requires a sequential-ignorability-type condition for the mediator-outcome relationship. In our setting, that means that once we condition on the remittance treatment, controls, and fixed effects, there should be no remaining unobserved factors that jointly affect the mediator and potential production outcomes. That is a strong requirement here. The mediators in (A.3) are all post-earthquake household responses, not pre-determined covariates. Households may differ in recovery needs, health shocks, local labor-market disruptions, expectations, or unobserved support from outside the formal remittance channel. Any of those factors could move both the mediator and production even if the total remittance effect is identified from exchange-rate-driven variation. This is why the total-effect IV design is more credible than the mediation decomposition itself.

We therefore interpret the mediation estimates as getting us closer to causal channel analysis, but not all the way there. The design is informative because each mediator is linked to the same triple-difference IV variation that identifies the main effect, and because the mediator patterns can be checked against the separate mechanism tables. Still, the strongest honest claim we can make is that Table 10 shows which post-earthquake margins are most strongly associated with the production effect within a disciplined IV decomposition. The mediation results are therefore best read together with the reduced-form evidence on consumption, capital, labor, and public assistance.

A.4 Shift-Share Analysis

The shift-share analysis measures a ward's predetermined exposure to migrant networks whose destination composition was already changing before the earthquake. For each ward w, let $z_{wc,2001}$ denote its baseline household exposure to destination group c. The destination groups are India, other SAARC countries, ASEAN, the Middle East, other Asia, the European Union, other Europe, North America, the Pacific, and other destinations including Africa and the Americas. The exposure is larger when a ward had stronger migrant links to destination groups that subsequently grew more rapidly elsewhere in Nepal. Formally,

$$SS_w = \sum_c z_{wc,2001} G_{c,-d,2001\text{–}11},$$

where $G_{c,-d,2001\text{–}11}$ is destination-specific migration growth between the two pre-earthquake census waves, calculated outside ward w's district d. Nepal had 75 districts during this period, so each ward's shifter is estimated from destination-specific migration growth in the other 74 districts. This leave-one-district-out growth factor prevents a ward's own local migration trajectory from mechanically determining its predicted exposure.

This construction follows the exposure logic of Khanna et al. (2026). The interpretation of a Bartik design depends on whether credibility is assigned to predetermined local shares (Goldsmith-Pinkham, Sorkin, and Swift 2020) or to quasi-random shocks (Borusyak, Hull, and Jaravel 2022). As emphasized in the practical shift-share guidance of Borusyak, Hull, and Jaravel (2025), these designs require a clear account of which variation is plausibly exogenous. In this setting, the design is most naturally interpreted as predetermined exposure to changing migrant networks rather than as a many-shock design. District-clustered standard errors also do not address arbitrary residual correlation across wards with similar exposure shares (Adão, Kolesár, and Morales 2019). We therefore treat the land-cover results as supporting evidence on the timing and production margin, not as a stand-alone migration IV.

The main-text specification uses indicator variables for high earthquake intensity and high shift-share migration exposure. Appendix Figure A1 and Appendix Table A8 show that the pattern is not driven by the binary shift-share cutoff. Instead, they retain the high-intensity earthquake indicator and enter shift-share exposure continuously. The continuous shift-share measure captures differences across wards in the strength of their pre-earthquake migrant links to destination groups whose migration networks were expanding elsewhere in Nepal, analogous to the origin-by-destination exposure logic in Khanna et al. (2026) and the destination-shock remittance logic in Yang (2008b). Each measure is standardized over the 2014 ward cross-section, so the triple interaction represents the differential effect of a one-standard-deviation increase in predicted migration exposure among high-earthquake-intensity wards.

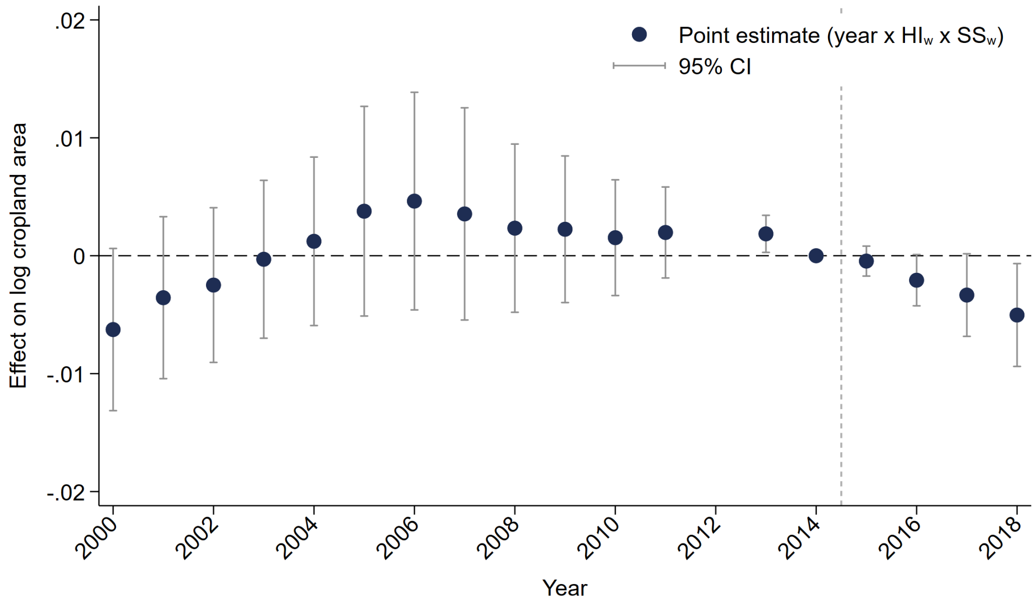

Figure A1 — Cropland event study with high earthquake intensity and continuous migration shift-share exposure

Note: The figure reports year-specific coefficients on year $\times HI_w \times SS_w$, where HI_w indicates above-median ward MMI and SS_w is continuous migration shift-share exposure, standardized over the 2014 ward cross-section. SS_w captures pre-earthquake ward exposure to migrant destination groups whose networks grew more rapidly in other districts before the earthquake. Destination groups include India, other SAARC countries, ASEAN, the Middle East, other Asia, Europe, North America, the Pacific, and other destinations. The omitted year is 2014. The dependent variable is $\log(C + 0.09)$, where C is ward cropland area in hectares. The specification includes ward fixed effects, district-year fixed effects, year-interacted core socioeconomic controls, and year interactions with the sum of baseline destination shares. Standard errors are clustered by district.

Table A8 — High-intensity and continuous migration shift-share exposure robustness

	Dependent variable: alternative transformations of cropland area C						
	$\log(C+0.09)$	ihs(C)	C (ha)	$\log(C)$	calibrated $\log(C)$		
				for $C>0$	x=0.1	x=0.5	x=1
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Pooled post-earthquake years 2015–18							
2015–18 x HI_w	0.004	0.003	0.982	0.004	0.003	0.003	0.003
	(0.006)	(0.006)	(0.683)	(0.006)	(0.006)	(0.006)	(0.006)
2015–18 x HI_w x SS_w	-0.003**	-0.003***	-0.174**	-0.004***	-0.003***	-0.003***	-0.003**
	(0.001)	(0.001)	(0.084)	(0.001)	(0.001)	(0.001)	(0.001)
Panel B. Separate post-earthquake years 2015–18							
2015 x HI_w x SS_w	-0.000	-0.001*	-0.068*	-0.001**	-0.001*	-0.001*	-0.001
	(0.001)	(0.001)	(0.037)	(0.001)	(0.001)	(0.001)	(0.001)
2016 x HI_w x SS_w	-0.002*	-0.003**	-0.165**	-0.003***	-0.003***	-0.003***	-0.003**
	(0.001)	(0.001)	(0.076)	(0.001)	(0.001)	(0.001)	(0.001)
2017 x HI_w x SS_w	-0.003*	-0.004**	-0.227**	-0.004**	-0.004**	-0.004**	-0.004**
	(0.002)	(0.002)	(0.109)	(0.002)	(0.002)	(0.002)	(0.002)
2018 x HI_w x SS_w	-0.005**	-0.006***	-0.236*	-0.006***	-0.006***	-0.006***	-0.005**
	(0.002)	(0.002)	(0.127)	(0.002)	(0.002)	(0.002)	(0.002)
N	617922	617922	617922	611598	617922	617922	617922
District	75	75	75	75	75	75	75
Ward and district-year FE	✓	✓	✓	✓	✓	✓	✓
Core controls	✓	✓	✓	✓	✓	✓	✓

Notes: This appendix table reports the high-intensity-by-continuous-shift-share robustness. HI_w indicates above-median ward MMI and SS_w is standardized continuous migration shift-share exposure. The shift-share combines pre-earthquake destination shares with leave-one-district-out destination growth across Nepal's 75 districts. C denotes ward cropland hectares. Both panels use the 2001–18 sample and omit 2014. Panel A pools the 2015–18 triple interaction; Panel B displays the separate 2015–18 triple interactions. All specifications include ward fixed effects, district-year fixed effects, year-interacted core socioeconomic controls, and year interactions with baseline destination-share totals. Standard errors are clustered by district. * p<0.10, ** p<0.05, *** p<0.01.

The continuous shift-share specification supports the same qualitative pattern. High-earthquake-intensity wards with stronger predicted migration exposure experience larger post-earthquake declines in cropland. The year-specific estimates become increasingly negative from 2015 through 2018. This pattern is similar across the logarithmic, inverse-hyperbolic-sine, and calibrated-log transformations, while the cropland level estimates are less precise.

The result is also similar when the exposure is expressed as predicted growth rather than the predicted level, which indicates that the pattern is not driven by baseline migration intensity alone.

A.5 Earthquake Intensity and Treatment Assignment

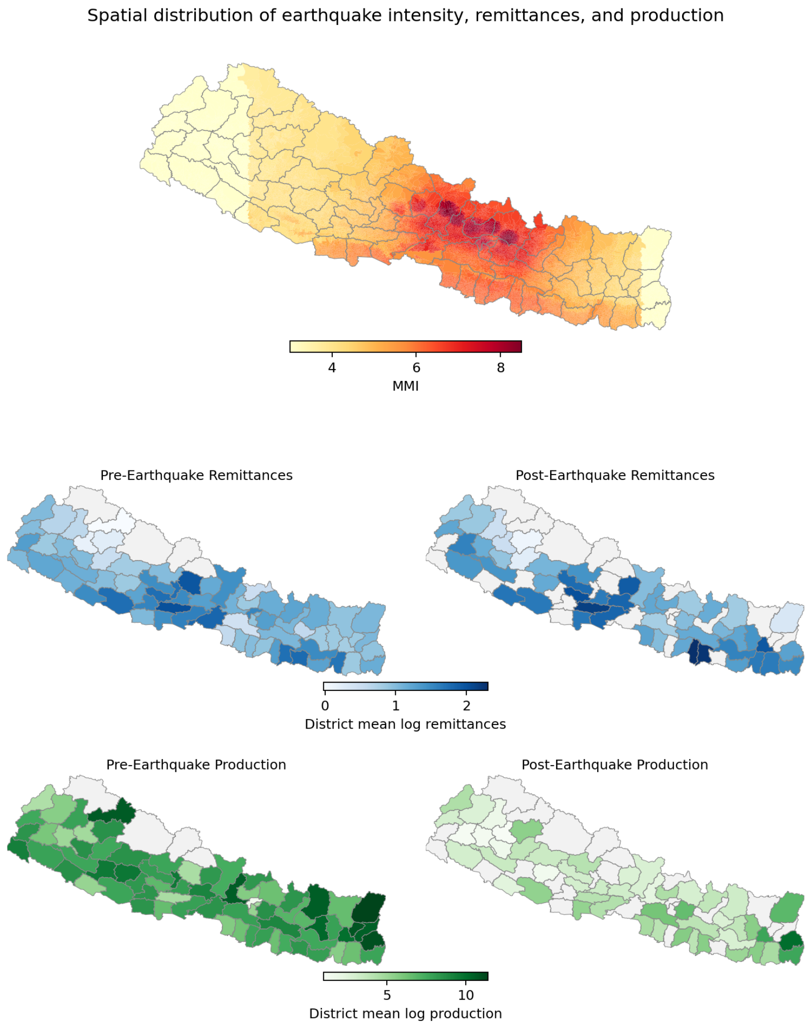

Figure A2 — Spatial distribution of earthquake intensity, remittances, and production

Note: The top panel shows ward-level earthquake intensity measured by the Modified Mercalli Intensity (MMI) scale. The lower panels show district-average pre- and post-earthquake remittances and production. Source: USGS, household-survey data, and authors' calculations.